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Outage performance analysis of antenna selection schemes in UAV-assisted networks

Ibrahim Al-Qarahghli [a], Khalid Yahya [a], Mahmoud Aldababsa [b,*]

[a] Electrical and Electronics Engineering Department, Istanbul Gelisim University, 34310, Istanbul, Turkey
[b] Electrical and Electronics Engineering Department, Nisantasi University, 34481742, Istanbul, Turkey

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ABSTRACT

In this paper, we investigate the performance of various antenna selection (AS) techniques in an unmanned aerial vehicle (UAV)-assisted relay communication system, wherein a UAV is deployed as a flying relay to establish and sustain a communication link between source and destination nodes. The AS schemes under study encompass transmit antenna selection/maximal ratio combining (TAS/MRC), joint transmit and receive antenna selection (JTRAS), and maximum ratio transmission/receive antenna selection (MRT/RAS). We present a unified outage probability (OP) performance analysis of AS techniques in UAV relay networks. We initially derive a closed-form expression for the OP under Nakagami-*m* fading channels. Subsequently, an asymptotic expression elucidating the array and diversity gain is derived to offer deeper insights into the performance of the studied system. Finally, Monte Carlo simulations are conducted to validate the theoretical results.

1. Introduction

Integrating Internet of Vehicles (IoV) communication into sixth-generation (6G) mobile networks has significantly facilitated big data transmission through IoV communication systems. It has paved the way for innovative 6G IoV applications, such as automatic driving, intent sharing, and interactive gaming. Current IoV communication systems face challenges in meeting transmission requirements, such as ultra-reliable and low-latency communication. The advent of 6G IoV has enabled broader signal coverage compared to 5G IoV, exceeding ground IoV capabilities. Additionally, given the escalating complexity of modern vehicular networks, characterized by diverse vehicles and dynamic topology, aerial platforms have gained prominence within the 6G IoV communication landscape. Unmanned aerial vehicles (UAVs) are extensively embraced to bolster wireless communications in IoV communication systems [1].

There are numerous applications for UAVs, including load capacity, flight altitude, and endurance time enhancements. Moreover, UAV communications are rapidly advancing and garnering significant attention from industry and academia. Contrary to traditional ground communications, UAV communications have several clear advantages. First, UAVs are easy to deploy and highly flexible. Equipped with communication devices, UAVs facilitate the swift establishment of communication links. The capability to regulate the UAV's altitude at any given moment enables adaptable coverage and network capacity, aligning with mission requirements, especially for time-sensitive missions. Due to their compact size, lightweight nature, and ease of transportation, UAVs can be flexibly deployed or redeployed to meet evolving business needs, reducing network costs and energy consumption. Secondly, UAV communications are not hindered by challenging terrain. Traditional wireless communications are frequently impeded by obstacles like tall buildings or high mountains, significantly deteriorating communication quality based on the height of the base station or ground devices. In contrast, UAVs possess aerial superiority, enabling them to circumvent obstacles and establish dependable communication links. Lastly, UAV equipment demonstrates robust applicability and high-quality information transmission. UAV platforms can readily upgrade communication equipment to enhance communication quality. In most cases, operating at low altitudes, UAVs enable the establishment of short-distance line-of-sight (LOS) communication links with ground users, significantly reducing path loss. These advantages position UAV communications as a promising technology for future wireless communication networks [2–4].

In the literature, UAVs, functioning as flying communication stations, present three typical scenarios to assist and enhance communications: UAV-aided relaying, UAV-aided ubiquitous coverage, and UAV-aided information dissemination and data collection. Notably, leveraging UAVs as aerial relays holds promise as an attractive solution to enhance the capabilities of existing wireless communications. Compared to traditional single-hop communications, relay transmission technology significantly extends the system's coverage, effectively

* Corresponding author.
E-mail address: mahmoud.aldababsa@nisantasi.edu.tr (M. Aldababsa).

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mitigates channel loss during information delivery, and bolsters the reliability of wireless communications. Given the advantages of UAVs, they emerge as promising candidates to serve as flexible flying relays, facilitating efficient data transmissions [5].

Multiple-input multiple-output (MIMO) technology has the potential to enhance the effectiveness of wireless systems concerning capacity and error rates. Antenna diversity methods are employed either at the transmission or reception ends of MIMO systems in various manners, mitigating the adverse impacts of fading channels in wireless communication. Despite yielding superior data capacity and enhancing overall system performance, these antenna diversity approaches contribute to heightened hardware complexity and increased power usage due to the presence of multiple radio frequency (RF) chains of antennas. However, antenna selection (AS) techniques, such as transmit antenna selection/maximal ratio combining (TAS/MRC) [6], joint Transmit and receive antenna selection (JTRAS) [7], and maximum ratio transmission/receive antenna selection (MRT/RAS) [8], offer a solution to mitigate the mentioned limitations while retaining the benefits of multi-antenna systems with a reduced count of RF chains [9,10].

1.1. Related work

In recent years, considerable research efforts have been directed toward the field of UAV relay communications. Jiang et al. [11] and Zeng et al. [12], in their respective studies, have developed throughput maximization algorithms that address the decode-and-forward (DF) and amplify-and-forward (AF) protocols. These algorithms jointly optimize both the trajectory of the UAV and the transmit power of the source/relay. Additionally, Zhang et al. [13] have explored the energy efficiency maximization when utilizing a UAV as a relay node for data exchange between a base station (BS) and a mobile device, focusing on the joint optimization of trajectory and power. A notable contribution comes from Zhang et al. [14], who successfully tackled the challenging problem of jointly optimizing trajectory and radio resources within cooperative UAV networks. Song and Zheng [15] considered a simplified scenario where a UAV follows a circular trajectory at a fixed altitude. In this setting, they jointly designed parameters such as beamforming, circle radius, flight speed, and power allocation to maximize energy efficiency. Furthermore, Wang et al. [16] examined a scenario involving full-duplex (FD) UAV relays and ground device-to-device pairs sharing the spectrum, which introduced the potential for mutual interference. Chen et al. [17] analyzed the optimal altitude for UAV relays, considering factors such as total power loss, total interruption probability, and total error rate to enhance reliability. Their study included a comparative evaluation of the performance of AF and DF relaying schemes under these varying reliability metrics. Dabiri and Sadough [18] derived optimal coordinates and beam patterns for UAV relays while examining the influence of factors like the intensity of the UAV's orientation fluctuations and the position and height of obstacles to address the minimization of outage probability. Takahashi et al. [19] and Ji et al. [20] delved into radio resource optimization schemes and energy harvesting schemes tailored explicitly for UAV relaying. Lastly, Song et al. [21] presented pioneering theoretical results that establish a correlation between the number of cooperative nodes and network capacity, which is paramount in designing collaborative communication networks, including those of interest in our UAV relay network investigations.

1.2. Motivations and contributions

The motivations of this paper with respect to the state-of-the-art literature can be highlighted as follows

- Most of the aforementioned studies on UAV communication networks have primarily focused on UAV-trajectory optimization [11,12,14,15,17,18], UAV-energy efficiency maximization [13,19,20], and UAV-performance analysis [16,21].
- Notably, a common aspect among these studies is the examination of UAV-assisted networks utilizing a single antenna. However, it is well established that employing multiple antennas has the potential to enhance the efficacy of wireless systems significantly.
- Consequently, this serves as a compelling impetus for our investigation into UAV-assisted networks employing multi-antenna configurations.

The distinct contributions of this paper are delineated as follows.

- We investigate the performance of various AS techniques, namely TAS/MRC, JTRAS, and MRT/RAS, within a UAV-assisted relay communication system. In this system, a UAV serves as a flying relay, facilitating establishing and maintaining communication links between source and destination nodes.
- We present a unified analysis of OP performance concerning the AS techniques employed in UAV relay networks. This analysis incorporates a unified assessment framework for the OP, consolidating insights into the reliability aspects of the system.
- Initially, we derive a closed-form expression characterizing the OP under Nakagami-m fading channels. This analytical foundation provides an essential basis for understanding reliability performance in challenging fading channel conditions.
- Furthermore, we conduct an asymptotic analysis in the regime of high signal-to-noise ratio (SNR), offering profound insights into the system's performance. By deriving a closed-form expression for the asymptotic OP, we gain substantial information regarding the array and diversity gains achieved by the system.
- Additionally, Monte Carlo simulations are conducted to validate and corroborate the theoretical findings. The simulation outcomes closely align with the derived theoretical expressions, affirming our analytical framework's accuracy. Notably, these results underscore a significant improvement in OP performance with an augmentation in transmit and receive antennas across all nodes. Moreover, heightened performance is observed as channel conditions improve. Specifically, the TAS/MRC and MRT/RAS schemes exhibit superior performance compared to JTRAS, albeit at the cost of increased computational complexity. Notably, the OP performance deteriorates with an increase in the UAV's distance from the ground nodes.

2. System and channel models

As shown in Fig. 1, two nodes—source (S) and destination (D)—are distributed on the ground. Communication over a direct path between S and D is not available due to a possible long distance between them, resulting in a significant path loss effect. Additionally, direct transmission is blocked by tremendous obstacles such as a building, or the direct path may suffer from high shadowing and/or severe fading effects. To enhance connectivity and enable real-time communication between S and D, a UAV (U) is dispatched as a flying relay. The U operates as a relay node and implements a DF relaying technique. Consequently, the S-to-D transmission process is divided into two time slots. First, the S transmits the message signal to the relay, and then the relay successfully decodes and forwards the message signal to D following the DF relaying technique. In this work, all nodes (i.e., S, U, and D) have multiple antennas (The S has N_1 transmit antennas, the U has N_2 receive and N_3 transmit antennas, and the D has N_4 receive antennas). We study various AS schemes. Specifically, we investigate the AS techniques (TAS/MRC, JTRAS, and MRT/RAS) in the first and second hops of the dual-hop.

Without loss of generality, we utilize a three-dimensional Cartesian coordinate system. The S and D are on the ground, and the U is fixed at the height h_U. Accordingly, the S, U, and D are located with the positions (x_S, y_S, h_S), (x_U, y_U, h_U), and (x_D, y_D, h_D), respectively, here

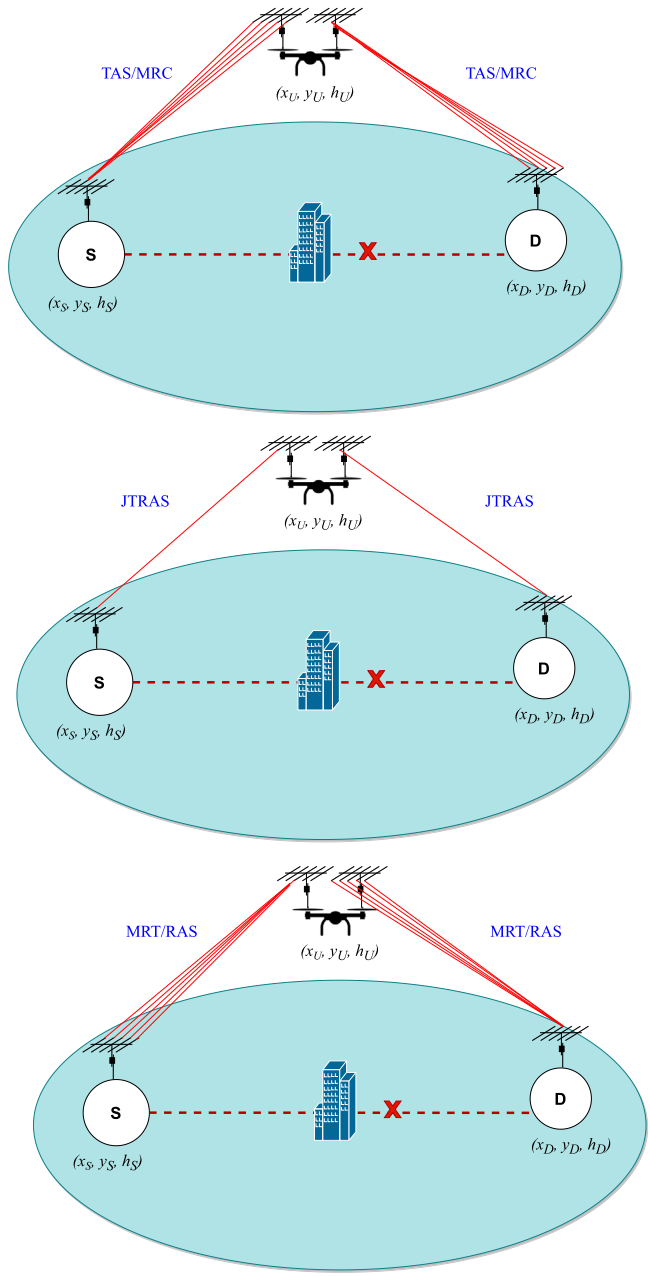

Fig. 1. The system model for UAV-assisted network with different AS schemes.

$h_S = h_D = 0$. The distances between the S and U (d_{SU}) and U and D (d_{UD}) are given, respectively, as

$$d_{SU} = \sqrt{(x_S - x_U)^2 + (y_S - y_U)^2 + (h_U)^2}, \tag{1}$$

and

$$d_{UD} = \sqrt{(x_U - x_D)^2 + (y_U - y_D)^2 + (h_U)^2}. \tag{2}$$

The links between the U and ground nodes (S and D) are mainly LoS or non-LoS (NLoS). The occurrence of LoS and NLoS transmissions can be captured using the probability of LoS transmission [22,23]. The expressions of the probability of LoS ($\mathcal{P}_{\varphi}^{los}$) and NLoS ($\mathcal{P}_{\varphi}^{nlos}$) can be given, respectively, as

$$\mathcal{P}_{\varphi}^{los} = \left(1 + \omega e^{-\varpi(\theta_{\varphi} - \omega)}\right)^{-1}, \tag{3}$$

and

$$\mathcal{P}_{\varphi}^{nlos} = 1 - \mathcal{P}_{\varphi}^{los}, \tag{4}$$

where $\varphi \in \{SU, UD\}$ denotes the link in the first hop between the S and U (SU) and the link in the second hop between the U and D (UD). Additionally, ω and ϖ refer to parameters contingent on different environments, such as rural, urban, or dense urban. Also, θ_{φ} denotes the elevation angle, which can be calculated as

$$\theta_{\varphi} = \frac{180}{\pi} \arcsin\left(\frac{h_U}{d_{\varphi}}\right) \tag{5}$$

Considering the probabilities of both LoS and NLoS links from the U to the ground node, given in (3) and (4), the mean path loss ($\mathcal{L}_{\varphi}$) can be stated as

$$\mathcal{L}_{\varphi} = \mathcal{L}_{\varphi}^{los}\mathcal{P}_{\varphi}^{los} + \mathcal{L}_{\varphi}^{nlos}\mathcal{P}_{\varphi}^{nlos}, \tag{6}$$

where $\mathcal{L}_{\varphi}^{los}$ and $\mathcal{L}_{\varphi}^{nlos}$ denote the loss exponent channels, which can be given, respectively, by

$$\mathcal{L}_{\varphi}^{los} = \frac{c\xi^{los}}{4\pi f_c d_{\varphi}^{\epsilon}}, \tag{7}$$

and

$$\mathcal{L}_{\varphi}^{nlos} = \frac{c\xi^{nlos}}{4\pi f_c d_{\varphi}^{\epsilon}}, \tag{8}$$

where c, f_c, and ϵ refer to the speed of light, carrier frequency and path loss exponent, respectively. In addition, ξ^{los} and ξ^{nlos} denote the excessive path losses of the LoS and NLoS propagation.

3. AS schemes

In this section, we investigate the following AS techniques in the first and second hops of the dual-hop: (1) TAS/MRC, (2) JTRAS, and (3) MRT/RAS.

3.1. TAS/MRC scheme

Based on the TAS/MRC scheme, a single transmit antenna that provides the maximum summation of channel gains is chosen for both the S and the U in the first and second hops, respectively [6]. Accordingly, the indices of the selected transmit antennas at the S (i_S^*) and U (i_U^*) can be stated, respectively, as

$$i_S^* = \underset{i=1,\ldots,N_1}{\arg\max}\left\{\sum_{j=1}^{N_2} |h_{SU}^{(i,j)}|^2\right\}, \tag{9}$$

and

$$i_U^* = \underset{i=1,\ldots,N_3}{\arg\max}\left\{\sum_{j=1}^{N_4} |h_{UD}^{(i,j)}|^2\right\}, \tag{10}$$

where $h_{SU}^{(i,j)}$ is the channel coefficient in the first hop between the ith transmit antenna of S and the jth receive antenna of U while $h_{UD}^{(i,j)}$ is the channel coefficient in the second hop between the ith transmit antenna of U and the jth receive antenna of D. Here, $|\cdot|$ refers to the absolute value.

The received signals at the U (y_U) and D (y_D) can be written, respectively, as

$$y_U = \sqrt{\frac{P_S}{\mathcal{L}_{SU}}}\mathbf{w}_{SU}\mathbf{h}_{SU}x^S + \mathbf{w}_{SU}\mathbf{n}_U, \tag{11}$$

and

$$y_D = \sqrt{\frac{P_U}{\mathcal{L}_{UD}}}\mathbf{w}_{UD}\mathbf{h}_{UD}x^U + \mathbf{w}_{UD}\mathbf{n}_D. \tag{12}$$

where P_S and P_U are the power transmission at the S and U, respectively, and x^S and x^U denote the signals at the S and U, respectively. In

addition, $\mathbf{h}_{SU}$ is the fading channel coefficients of the first hop between the selected transmit antenna i_S^* of S and U while $\mathbf{h}_{UD}$ is the fading channel coefficients of the second hop between the selected transmit antenna i_U^* of U and D. Also, $\mathbf{w}_{SU}$ and $\mathbf{w}_{UD}$ are the optimal receive weight vectors at the U and D, respectively, with $\mathbf{w}_{SU} = \frac{\mathbf{h}_{SU}^H}{\|\mathbf{h}_{SU}\|}$ and $\mathbf{w}_{UD} = \frac{\mathbf{h}_{UD}^H}{\|\mathbf{h}_{UD}\|}$. $\mathbf{n}_U$ and $\mathbf{n}_D$ are zero mean complex additive Gaussian noise with σ^2, respectively at the U and D. Here, $(\cdot)^H$ denotes the Hermitian operator and $\|\cdot\|$ refers to the Norm value.

3.2. JTRAS scheme

Based on the JTRAS scheme, the transmit-receive antenna pair between the S and U, (i_S^*, j_U^*) is chosen in the first hop while the transmit-receive antenna pair between the U and D, (i_U^*, j_D^*) is selected in the second hop [7]. Precisely, (i_S^*, j_U^*) has the highest channel gain between the transmit antennas of S and receiver antennas of the U and it can be expressed as

$$(i_S^*, j_U^*) = \underset{\substack{i=1,\ldots,N_1\\ j=1,\ldots,N_2}}{\arg\max}\{|h_{SU}^{(i,j)}|^2\}, \tag{13}$$

On the other hand, (i_U^*, j_D^*) has the maximum channel gain between the transmit antennas of U and receiver antennas of the D and it can be written as

$$(i_U^*, j_D^*) = \underset{\substack{i=1,\ldots,N_3\\ j=1,\ldots,N_4}}{\arg\max}\{|h_{UD}^{(i,j)}|^2\}. \tag{14}$$

The received signal of U and D can be respectively expressed as

$$y_U = \sqrt{\frac{P_S}{\mathcal{L}_{SU}}} h_{SU} x^S + n_U, \tag{15}$$

and

$$y_D = \sqrt{\frac{P_U}{\mathcal{L}_{UD}}} h_{UD} x^U + n_D, \tag{16}$$

where h_{SU} is the channel coefficient corresponding to the best link in the first hop, offering the highest channel gain between the S and U while h_{UD} is the channel coefficient corresponding to the best link in the second hop, offering the highest channel gain between the U and D. Additionally, n_U and n_D refer to the zero mean complex additive Gaussian noise with a variance of σ^2, respectively at the U and D.

3.3. MRT/RAS scheme

Based on the MRT/RAS scheme, single receive antennas that offer the maximum summation of channel gains are chosen for the U and D in the first and second hops, respectively [8]. Accordingly, the indices of the chosen receive antennas at the U (j_U^*) and D (j_D^*) can be stated, respectively, as

$$j_U^* = \underset{j=1,\ldots,N_2}{\arg\max}\{\sum_{i=1}^{N_1} |h_{SU}^{(i,j)}|^2\}, \tag{17}$$

and

$$j_D^* = \underset{j=1,\ldots,N_4}{\arg\max}\{\sum_{i=1}^{N_3} |h_{UD}^{(i,j)}|^2\}. \tag{18}$$

The received signals at the U and D can be written, respectively, as

$$y_U = \sqrt{\frac{P_S}{\mathcal{L}_{SU}}} \mathbf{h}_{SU} \mathbf{u}_{SU} x^S + n_U, \tag{19}$$

and

$$y_D = \sqrt{\frac{P_U}{\mathcal{L}_{UD}}} \mathbf{h}_{UD} \mathbf{u}_{UD} x^U + n_D, \tag{20}$$

where $\mathbf{h}_{SU}$ is the fading channel coefficients of the first hop between the selected receive antenna j_U^* of U and S while $\mathbf{h}_{UD}$ is the fading channel coefficients of the second hop between the selected receive antenna j_D^* of D and U. Additionally, $\mathbf{h}_{UD}$ $\mathbf{u}_{SU}$ and $\mathbf{u}_{UD}$ are the optimal transmit weight vectors at the S and U, respectively, with $\mathbf{u}_{SU} = \frac{\mathbf{h}_{SU}^H}{\|\mathbf{h}_{SU}\|}$ and $\mathbf{u}_{UD} = \frac{\mathbf{h}_{UD}^H}{\|\mathbf{h}_{UD}\|}$.

4. OP analysis

Considering the received signals at the U and D for various AS schemes ((11), (12), (15), (16), (19), and (20)) and DF relaying, the unified expression for end-to-end (e2e) SNR (γ_{SD}^{e2e}) can be written as

$$\gamma_{SD}^{\text{e2e}} = \min\left(\gamma_{SU}, \gamma_{UD}\right), \tag{21}$$

where γ_{SD} and γ_{UD} are the SNR values at the U and the D, respectively. The expressions of γ_{SD} and γ_{UD} for AS techniques are given in Table 1. Here, we assume $\gamma = \frac{P_S}{\sigma^2} = \frac{P_U}{\sigma^2}$,[1] which is the average SNR.

Now, the OP can be calculated as

$$\begin{aligned}\text{OP} &= F_{\gamma_{SD}^{\text{e2e}}}(x)\\ &= \mathcal{P}\left(\gamma_{SD}^{\text{e2e}} \le x\right),\end{aligned} \tag{22}$$

where $\mathcal{P}(\cdot)$ denotes the probability operator and $x = \frac{\gamma_{th}\mathcal{L}_\varphi}{\gamma}$. Here, γ_{th} presents the threshold SNR value. Based on the expression of γ_{SD}^{e2e} in (21), the OP in (22) can be rewritten as

$$\begin{aligned}\text{OP} &= \mathcal{P}\left(\min\left(\gamma_{SU}, \gamma_{UD}\right) \le x\right)\\ &= 1 - \left(1 - F_{\gamma_{SU}}(x)\right)\left(1 - F_{\gamma_{UD}}(x)\right)\\ &= F_{\gamma_{SU}}(x) + F_{\gamma_{UD}}(x) - F_{\gamma_{SU}}(x) F_{\gamma_{UD}}(x).\end{aligned} \tag{23}$$

In the following, the expressions of $F_{\gamma_{SU}}(x)$ and $F_{\gamma_{UD}}(x)$ in (23) are derived.

It is worthing that the channel gains between S and U for the first hop and U and D for the second hop are assumed to follow the Nakagami-m fading model. The cumulative distribution function (CDF) of the squared channel gain between the links in the first and second hops can be stated as [9,10]

$$F_{|h_\varphi^{(i,j)}|^2}(x) = \frac{\Psi(m_\varphi, \frac{m_\varphi x}{\Omega_\varphi})}{\Gamma(m_\varphi)}, \tag{24}$$

where Ω_φ and m_φ are corresponding parameter of Nakagami fading.

Considering the expressions of γ_{SD} and γ_{UD} for AS techniques in Table 1 and basing on highest-order statistics, the unified expression for the CDF of γ_{SD} and γ_{UD} can be written as [9,10]

$$F_{\gamma_\varphi}(x) = \left(\frac{\gamma(\alpha_\varphi, \frac{m_\varphi x}{\Omega_\varphi})}{\Gamma(\alpha_\varphi)}\right)^{\beta_\varphi}. \tag{25}$$

Here the values of α_φ and β_φ for different AS schemes are given in Table 2.

To simplify the expression in (25): first, by utilizing an alternative expression of lower incomplete gamma functions, the CDF in (25) can be rewritten as

$$F_{\gamma_\varphi}(x) = \left(1 - e^{-\frac{m_\varphi x}{\Omega_\varphi}} \sum_{k=0}^{\alpha_\varphi - 1} \left(\frac{m_\varphi x}{\Omega_\varphi}\right)^k \frac{1}{k!}\right)^{\beta_\varphi}. \tag{26}$$

[1] In many practical situations, this assumption, $P_S = P_U$, may not hold true due to various factors. However, this assumption is made in this paper to simplify the analysis and provide a baseline scenario for comparison.

Table 1
Expressions of γ_{SD} and γ_{UD} for AS techniques.

AS scheme	γ_{SU}	γ_{UD}
TAS/MRC	$\frac{\gamma}{\mathcal{L}_{SU}} \parallel \mathbf{h}_{SU} \parallel^2 = \frac{\gamma}{\mathcal{L}_{SU}} \max_{i=1,\ldots,N_1} \left\{ \sum_{j=1}^{N_2} \lvert h_{SU}^{(i,j)} \rvert^2 \right\}$	$\frac{\gamma}{\mathcal{L}_{UD}} \parallel \mathbf{h}_{UD} \parallel^2 = \frac{\gamma}{\mathcal{L}_{UD}} \max_{i=1,\ldots,N_3} \left\{ \sum_{j=1}^{N_4} \lvert h_{UD}^{(i,j)} \rvert^2 \right\}$
JTRAS	$\frac{\gamma}{\mathcal{L}_{SU}} \lvert h_{SU} \rvert^2 = \frac{\gamma}{\mathcal{L}_{SU}} \max_{\substack{i=1,\ldots,N_1 \\ j=1,\ldots,N_2}} \left\{ \lvert h_{SU}^{(i,j)} \rvert^2 \right\}$	$\frac{\gamma}{\mathcal{L}_{UD}} \lvert h_{UD} \rvert^2 = \frac{\gamma}{\mathcal{L}_{UD}} \max_{\substack{i=1,\ldots,N_3 \\ j=1,\ldots,N_4}} \left\{ \lvert h_{UD}^{(i,j)} \rvert^2 \right\}$
MRT/RAS	$\frac{\gamma}{\mathcal{L}_{SU}} \parallel \mathbf{h}_{SU} \parallel^2 = \frac{\gamma}{\mathcal{L}_{SU}} \max_{j=1,\ldots,N_2} \left\{ \sum_{i=1}^{N_1} \lvert h_{SU}^{(i,j)} \rvert^2 \right\}$	$\frac{\gamma}{\mathcal{L}_{UD}} \parallel \mathbf{h}_{UD} \parallel^2 = \frac{\gamma}{\mathcal{L}_{UD}} \max_{j=1,\ldots,N_4} \left\{ \sum_{i=1}^{N_3} \lvert h_{UD}^{(i,j)} \rvert^2 \right\}$

Table 2
Values of α_φ and β_φ for different AS schemes.

AS scheme	α	β
TAS/MRC	$\alpha_{SU} = m_{SU} N_2$ and $\alpha_{UD} = m_{UD} N_4$	$\beta_{SU} = N_1$ and $\beta_{UD} = N_3$
JTRAS	$\alpha_{SU} = m_{SU}$ and $\alpha_{UD} = m_{UD}$	$\beta_{SU} = N_1 N_2$ and $\beta_{UD} = N_3 N_4$
MRT/RAS	$\alpha_{SU} = m_{SU} N_1$ and $\alpha_{UD} = m_{UD} N_3$	$\beta_{SU} = N_2$ and $\beta_{UD} = N_4$

Second, with the help of multinomial coefficients, the CDF in (26) can be expressed as

$$F_{\gamma_\varphi}(x) = \sum_{p=0}^{\beta_\varphi} \sum_{k=0}^{p(\alpha_\varphi - 1)} \binom{\beta_\varphi}{p} (-1)^p \vartheta_k(p, \alpha_\varphi) x^k e^{-\frac{p m_\varphi x}{\Omega_\varphi}}, \tag{27}$$

where $\vartheta_a(b, c)$ denotes multinomial coefficients, which can be written as [24]

$$\vartheta_a(b, c) = \frac{1}{a e_0} \sum_{d=1}^{a} (d(b+1) - a) e_d \vartheta_{a-b}(b, c), a \geq 1, \tag{28}$$

where $e_d = (c/f)^d / d!$, $\vartheta_0(b, c) = 1$, and $\vartheta_a(b, c) = 0$ if $d > c - 1$. Finally, by utilizing (27) to obtain the expressions of $F_{\gamma_{SU}}(x)$ and $F_{\gamma_{UD}}(x)$ and substituting them into (23), the closed-form expression of the OP can be written as in (29).

$$\begin{aligned} \mathrm{OP} = & \sum_{p_1=0}^{\beta_{SU}} \sum_{k_1=0}^{p_1(\alpha_{SU}-1)} \binom{\beta_{SU}}{p_1} (-1)^{p_1} \vartheta_{k_1}(p_1, \alpha_{SU}) x^{k_1} e^{-\frac{p_1 m_{SU} x}{\Omega_{SU}}} \\ & + \sum_{p_2=0}^{\beta_{UD}} \sum_{k_2=0}^{p_2(\alpha_{UD}-1)} \binom{\beta_{UD}}{p_2} (-1)^{p_2} \vartheta_{k_2}(p_2, \alpha_{UD}) x^{k_2} e^{-\frac{p_2 m_{UD} x}{\Omega_{UD}}} \\ & - \sum_{p_1=0}^{\beta_{SU}} \sum_{k_1=0}^{p_1(\alpha_{SU}-1)} \sum_{p_2=0}^{\beta_{UD}} \sum_{k_2=0}^{p_2(\alpha_{UD}-1)} \binom{\beta_{SU}}{p_1} \binom{\beta_{UD}}{p_2} (-1)^{p_1+p_2} \\ & \times \vartheta_{k_1}(p_1, \alpha_{SU}) \vartheta_{k_2}(p_2, \alpha_{UD}) x^{k_1+k_2} e^{-\frac{p_1 m_{SU} x}{\Omega_{SU}} - \frac{p_2 m_{UD} x}{\Omega_{UD}}} \end{aligned} \tag{29}$$

5. Asymptotic OP

Recall, $F_{\gamma_{SU}}(x) F_{\gamma_{UD}}(x)$ represents the product of two CDFs, likely associated with random variables γ_{SU} and γ_{UD}. In certain mathematical contexts, when multiplying two CDFs, the resulting value can be considerably smaller compared to other terms in the equation. In other words, the value of $F_{\gamma_{SU}}(x) F_{\gamma_{UD}}(x)$ is too small, and then this term can be neglected in (23). Accordingly, the asymptotic OP can be calculated as

$$\mathrm{OP}^\infty = F_{\gamma_{SU}}^\infty(x) + F_{\gamma_{UD}}^\infty(x). \tag{30}$$

Now, using a high SNR approximation technique [25] and the property of incomplete Gamma function [26], the expression of the asymptotic CDF functions ($F_{\gamma_{SU}}^\infty(x)$ and $F_{\gamma_{UD}}^\infty(x)$) can be expressed generally as

$$F_{\gamma_\varphi}^\infty(x) = \frac{\left(\frac{m_\varphi}{\Omega_\varphi}\right)^{\alpha_\varphi \beta_\varphi}}{\Gamma(\alpha_\varphi + 1)^{\beta_\varphi}} x^{\alpha_\varphi \beta_\varphi}. \tag{31}$$

By substituting $F_{\gamma_{SU}}^\infty(x)$ and $F_{\gamma_{UD}}^\infty(x)$) from (31) into (30), then the asymptotic OP can be expressed as

$$\mathrm{OP}^\infty = \frac{\left(\frac{m_{SU}}{\Omega_{SU}}\right)^{\alpha_{SU} \beta_{SU}}}{\Gamma(\alpha_{SU} + 1)^{\beta_{SU}}} x^{\alpha_{SU} \beta_{SU}} + \frac{\left(\frac{m_{UD}}{\Omega_{UD}}\right)^{\alpha_{UD} \beta_{UD}}}{\Gamma(\alpha_{UD} + 1)^{\beta_{UD}}} x^{\alpha_{UD} \beta_{UD}}. \tag{32}$$

Table 3
Parameters used in the simulation.

Parameter	Value
(x_S, y_S, h_S)	(0, 0, 0) m
(x_U, y_U, h_U)	(100, 0, 1000) m
(x_D, y_D, h_D)	(200, 0, 0) m
ω	0.1581
ϖ	9.6117
c	3×10^8 m/s
f_c	100×10^3 Hz/s
ϵ	3
ξ^{los}	1
ξ^{nlos}	20
Ω_{SU}	1
Ω_{UD}	1

From (32), the asymptotic OP can be re-expressed as

$$\mathrm{OP}^\infty = (G^a x)^{G^d}, \tag{33}$$

where G^d and G^a denote the diversity and array gains, respectively, and are expressed in (34) and (35).

$$G^d = \min\left(\alpha_{SU} \beta_{SU}, \alpha_{UD} \beta_{UD}\right). \tag{34}$$

$$G^a = \begin{cases} \dfrac{\left(\frac{m_{SU}}{\Omega_{SU}}\right)^{\alpha_{SU} \beta_{SU}}}{\Gamma(\alpha_{SU} + 1)^{\beta_{SU}}}, & \alpha_{SU} \beta_{SU} < \alpha_{UD} \beta_{UD} \\ \dfrac{\left(\frac{m_{SU}}{\Omega_{SU}}\right)^{\alpha_{SU} \beta_{SU}}}{\Gamma(\alpha_{SU} + 1)^{\beta_{SU}}} + \dfrac{\left(\frac{m_{UD}}{\Omega_{UD}}\right)^{\alpha_{UD} \beta_{UD}}}{\Gamma(\alpha_{UD} + 1)^{\beta_{UD}}}, & \alpha_{SU} \beta_{SU} = \alpha_{UD} \beta_{UD} \\ \dfrac{\left(\frac{m_{UD}}{\Omega_{UD}}\right)^{\alpha_{UD} \beta_{UD}}}{\Gamma(\alpha_{UD} + 1)^{\beta_{UD}}}, & \alpha_{SU} \beta_{SU} > \alpha_{UD} \beta_{UD} \end{cases}. \tag{35}$$

6. Numerical results

In this section, we present numerical results to investigate the performance of the considered system. The simulation parameter values are given in Table 3.

Fig. 2 plots the OP performance of the TAS/MRC scheme in the UAV-assisted network as a function of the average SNR and for different fading parameters ($m = m_{SU} = m_{UD}$) and antenna configurations (N_1, N_2, N_3, N_4). We have the following observations: the analytical and simulation results match perfectly as well as the asymptotic OP results converge to the analytical and simulation results at very high values of SNR. Moreover, the OP performance improves significantly with

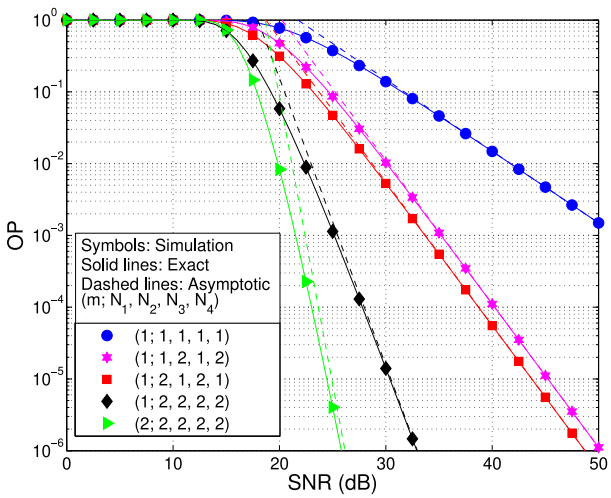

Fig. 2. The OP performance for the TAS/MRC scheme in the UAV-assisted network.

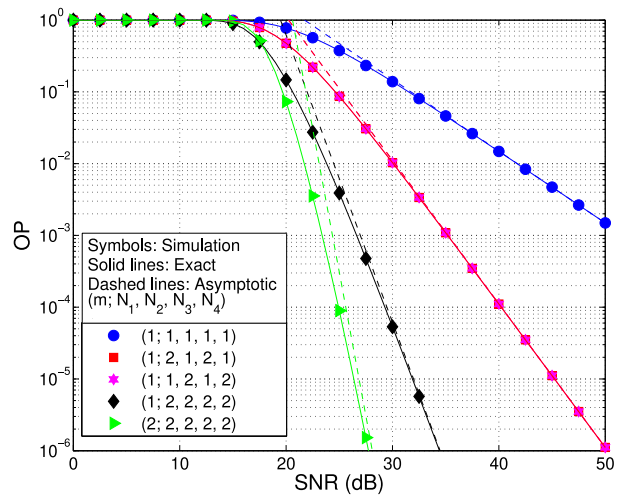

Fig. 3. The OP performance for the JTRAS scheme in the UAV-assisted network.

increased transmit and receive antennas at all nodes. Specifically, we utilize $(N_1, N_2, N_3, N_4) = (1, 1, 1, 1)$, $(2, 1, 2, 1)$, $(1, 2, 1, 2)$, and $(2, 2, 2, 2)$ to represent the cases of single-antenna, TAS/1, 1/MRC, and TAS/MRC, respectively. Clearly, the TAS/MRC offers a significant enhancement in performance compared to other cases. For example, the configuration $(N_1, N_2, N_3, N_4) = (2, 2, 2, 2)$ in the case $m = 1$ provides about 19 dB, 7 dB and 5 dB SNR gains at an OP of 10^{-2} as compared to $(1, 1, 1, 1)$, $(2, 1, 2, 1)$, and $(1, 2, 1, 2)$, respectively. Furthermore, the OP performance is enhanced when the channel condition improves. Particularly, we use $m = 1$ and $m = 2$ to represent the fading channel as a special case (Rayleigh fading) and Nakagami-m fading, respectively. For example, the configuration $(N_1, N_2, N_3, N_4) = (2, 2, 2, 2)$ in the case of $m = 2$ provides approximately 5 dB SNR gain at an OP of 10^{-5} compared to $(2, 2, 2, 2)$ in the case of $m = 1$.

Fig. 3 shows the OP performance of the JTRAS scheme in the UAV-assisted network versus the average SNR for various fading parameters and antenna configurations. It is evident that the OP performance improves considerably with an increased number of transmit and receive antennas at all nodes. Specifically, we utilize $(N_1, N_2, N_3, N_4) = (1, 1, 1, 1)$, $(2, 1, 2, 1)$, $(1, 2, 1, 2)$, and $(2, 2, 2, 2)$ to represent the cases of a single-antenna, TAS/1, 1/RAS, and JTRAS, respectively. Clearly, JTRAS outperforms the other cases. For instance, the configuration $(N_1, N_2, N_3, N_4) = (2, 2, 2, 2)$ in the case $m = 1$ provides about 17 dB, 5 dB, and 5 dB SNR gains at an OP of 10^{-2} compared to $(1, 1, 1, 1)$, $(2, 1, 2, 1)$, and $(1, 2, 1, 2)$, respectively. Furthermore, the OP performance is also enhanced as the channel condition improves. For example, the configuration $(N_1, N_2, N_3, N_4) = (2, 2, 2, 2)$ in the case of $m = 2$ provides approximately 4 dB SNR gain at an OP of 10^{-5} compared to $(2, 2, 2, 2)$ in the case of $m = 1$.

Fig. 4 demonstrates the OP performance of the MRT/RAS scheme in the UAV-assisted network versus the average SNR for various fading parameters and antenna configurations. We can observe the enhancement in OP performance with increased transmit and receive antennas at all nodes and with improved channel conditions. Specifically, we utilize $(N_1, N_2, N_3, N_4) = (1, 1, 1, 1)$, $(2, 1, 2, 1)$, $(1, 2, 1, 2)$, and $(2, 2, 2, 2)$ to represent the cases of a single-antenna, MRT/1, 1/RAS, and MRT/RAS, respectively. MRT/RAS surpasses the other cases. For instance, the configuration $(N_1, N_2, N_3, N_4) = (2, 2, 2, 2)$ in the case $m = 1$ provides about 19 dB,7 dB, and 5 dB SNR gains at an OP of 10^{-2} compared to $(1, 1, 1, 1)$, $(2, 1, 2, 1)$, and $(1, 2, 1, 2)$, respectively. On the other hand, the configuration $(N_1, N_2, N_3, N_4) = (2, 2, 2, 2)$ in the case of $m = 2$ provides approximately 5 dB SNR gain at an OP of 10^{-5} compared to $(2, 2, 2, 2)$ in the case of $m = 1$.

Fig. 5 presents a comparison between AS schemes (TAS/MRC, JTRAS and MRT/RAS). According to the results, TAS/MRC and MRT/RAS

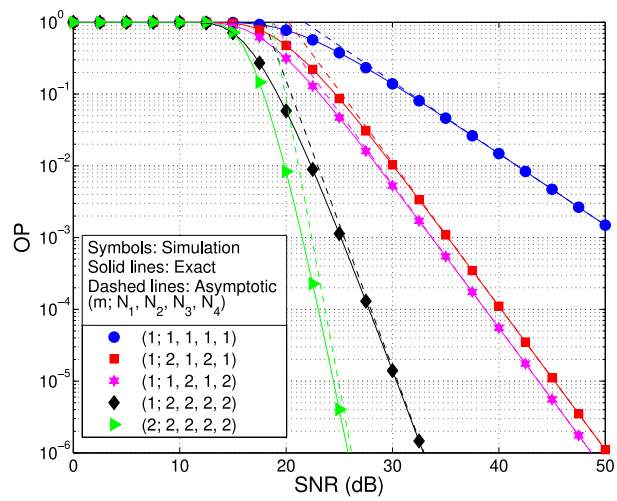

Fig. 4. The OP performance for the MRT/RAS scheme in the UAV-assisted network.

demonstrate similar optimal performance compared to JTRAS. In TAS/MRC, the relay actively utilizes all antennas for reception, but only one antenna is used for transmission. On the other hand, in MRT/RAS, the source actively utilizes all antennas for transmission but only a single antenna for reception. In JTRAS, all nodes use a single antenna actively. Therefore, TAS/MRC and MRT/RAS perform better than JTRAS. However, it is important to note that JTRAS reduces complexity since it uses only a single RF chain, while TAS/MRC and MRT/RAS use N RF chains, assuming all nodes are equipped with N antennas.

Fig. 6 shows the behavior of the system when the UAV moves away from either the source or destination. To comprehend this phenomenon, it is important to observe that as the distance from the source increases, the likelihood of an outage for the first transmission link rises. Likely, as the distance from the destination increases, the probability of outage for the second transmission link also increases, and vice versa. As Fig. H illustrated, the optimal UAV height, resulting in the lowest OP, is 100 m across all antenna configuration cases (single antenna, JTRAS, TAS/MRC, and MRT/RAS). Furthermore, it is noteworthy that the OP approaches a value of 1 for all antenna schemes (single antenna, JTRAS, TAS/MRC, and MRT/RAS) as the SNR decreases. This trend can be attributed to the UAV being positioned farther away, resulting in weaker signal reception and thus a higher likelihood of communication outages.

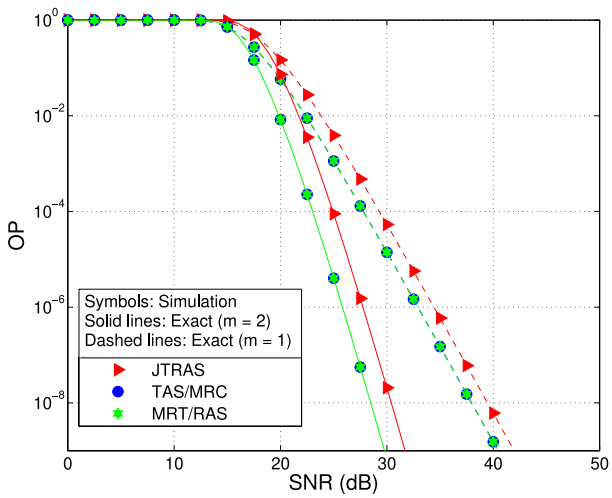

Fig. 5. The comparison between the AS schemes.

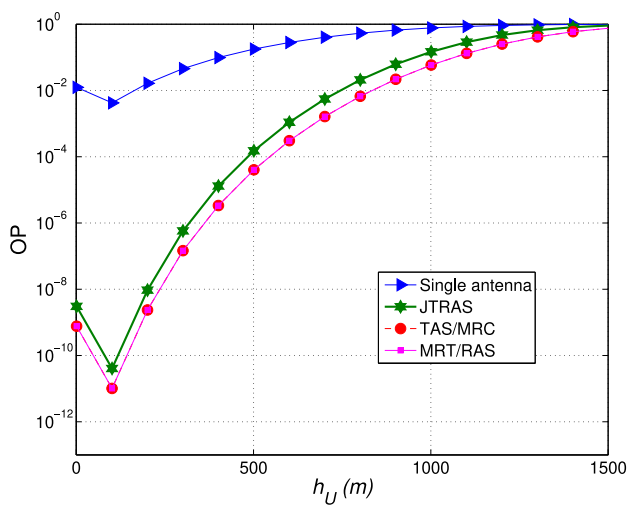

Fig. 6. The OP versus height of UAV for different AS schemes.

7. Conclusion

In this study, we have thoroughly investigated and compared the performance of various AS techniques in a relay communication system employing UAV as a flying relay. The AS schemes studied included TAS/MRC, JTRAS, and MRT/RAS. Our analysis encompassed the derivation of a closed-form expression for the OP under Nakagami-*m* fading channels, providing fundamental insights into the system's reliability. Furthermore, we derived an asymptotic expression characterizing the array and diversity gain, shedding light on the potential improvements in system performance with different AS techniques. These analytical results offer valuable theoretical foundations for understanding the behavior of AS in UAV relay networks and aid in designing efficient communication systems. To validate our theoretical findings, Monte Carlo simulations were conducted. The simulation results align closely with the derived theoretical expressions, affirming the accuracy and applicability of our analytical framework. Moreover, the results show that the OP performance improves considerably with increased transmit and receive antennas at all nodes. The OP performance is also enhanced as the channel condition improves. Furthermore, the TAS/MRC and MRT/RAS schemes perform better than JTRAS, while JTRAS reduces complexity. Finally, the OP performance becomes worse as the UAV becomes far away.

CRediT authorship contribution statement

Ibrahim Al-Qarahghli: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Khalid Yahya:** Writing – review & editing, Writing – original draft. **Mahmoud Aldababsa:** Conceptualization, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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